
Remote Direct Memory Access

& RDMA over Converged Ethernet (RoCE)

A Technical Overview of High-Performance Network Memory Access

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Abstract

Remote Direct Memory Access (RDMA) is a network technology that allows a computer to access the memory of a remote machine without involving the remote CPU, operating system, or cache. This capability enables extremely low-latency, high-throughput data transfers that are critical to modern high-performance computing (HPC), cloud data centers, and distributed storage systems. This report provides a comprehensive technical overview of RDMA, including its architectural foundations, key transport protocols, the RDMA over Converged Ethernet (RoCE) standard, performance considerations, and real-world deployment patterns.

1 Introduction

As data center workloads grow in scale and complexity, the bottleneck has shifted from compute to networking. Traditional networking stacks require the CPU to process every packet — copying data between kernel and user space, handling interrupts, and managing socket buffers. This overhead becomes prohibitive in latency-sensitive, bandwidth-intensive applications such as distributed machine learning, in-memory databases, and high-frequency trading.

RDMA addresses this bottleneck by implementing a *kernel-bypass* mechanism: data is moved directly between application memory regions across a network, with the network interface card (NIC) handling the transfer autonomously. The result is microsecond-level latencies and throughputs that approach the theoretical limits of the underlying physical medium.

1.1 Historical Context

RDMA was pioneered through the **InfiniBand** architecture, introduced in 2000 as a replacement for the PCI bus and later adopted for cluster interconnects. InfiniBand provided native RDMA support in its protocol stack and became the dominant fabric in Top500 supercomputers. However, its proprietary nature and separate cabling infrastructure led the industry to seek RDMA capabilities over standard Ethernet, culminating in the development of RoCE.

1.2 Motivation and Use Cases

Key use cases for RDMA include:

- **High-Performance Computing (HPC):** MPI (Message Passing Interface) libraries use RDMA for inter-node communication in scientific simulations.
- **Distributed Machine Learning:** Gradient synchronization in distributed training (e.g., NCCL, collective operations) benefits enormously from RDMA's low latency.
- **In-Memory Databases:** Systems like VoltDB and Redis cluster editions use RDMA to replicate state and serve queries with minimal overhead.
- **Network-Attached Storage (NAS/SAN):** NVMe over Fabrics (NVMe-oF) uses RDMA transports (RoCE, iWARP) for remote block storage.
- **Cloud Data Centers:** Microsoft Azure, Google, and AWS leverage RoCE internally for VM live migration, storage backends, and AI training clusters.

2 RDMA Architecture

2.1 Core Concepts

RDMA is built around several key abstractions:

Queue Pair (QP)

The fundamental communication endpoint in RDMA. A QP consists of a *Send Queue* (SQ) and a *Receive Queue* (RQ). Operations are posted to these queues by the application.

Completion Queue (CQ)

Holds completion notifications for operations posted to Queue Pairs. Applications poll the CQ to determine when operations are done.

Memory Region (MR)

A registered region of virtual memory that the RDMA NIC (RNIC) is permitted to access. Registration pins pages in physical memory and maps them into the NIC's address space.

Protection Domain (PD)

A security container that groups QPs and MRs together, preventing unauthorized cross-process memory access.

Address Handle (AH)

Encapsulates the routing information needed to reach a remote endpoint (used in unreliable datagram operations).

2.2 RDMA Verbs API

Applications interact with RDMA hardware through the **Verbs API**, originally defined by the InfiniBand specification and implemented by the `libibverbs` library on Linux. The API exposes primitives such as:

Listing 1: Key RDMA Verbs API calls

```

1 // Register memory region
2 struct ibv_mr *ibv_reg_mr(struct ibv_pd *pd, void *addr,
3                           size_t length, int access);
4
5 // Post a send operation
6 int ibv_post_send(struct ibv_qp *qp,
7                  struct ibv_send_wr *wr,
8                  struct ibv_send_wr **bad_wr);
9
10 // Poll completion queue
11 int ibv_poll_cq(struct ibv_cq *cq, int num_entries,
12                struct ibv_wc *wc);

```

Operation	Description	CPU involvement	Direction
SEND / RECV	Two-sided message passing	Both sides	Bidirectional
RDMA WRITE	Write to remote memory	Initiator only	Initiator → Target
RDMA READ	Read from remote memory	Initiator only	Target → Initiator
ATOMIC	Fetch-and-Add, Compare-and-Swap	Initiator only	Bidirectional

Table 1: RDMA operation types and CPU involvement

2.3 RDMA Operation Types

RDMA supports several distinct operation types, each with different semantics:

The defining characteristic of RDMA WRITE and RDMA READ is **one-sided operation**: the remote CPU is not involved and does not need to be notified. This is the key enabler of RDMA’s latency advantages.

3 Transport Protocols

Three primary transport protocols carry RDMA traffic in production environments:

3.1 InfiniBand (IB)

InfiniBand is a purpose-built, lossless interconnect that provides native RDMA support. It uses a dedicated physical layer with its own switch fabric (managed by a Subnet Manager) and supports link speeds up to 400 Gb/s (HDR200) and beyond (NDR: 400 Gb/s per port). InfiniBand dominates in HPC but requires dedicated hardware and cabling.

3.2 iWARP

Internet Wide Area RDMA Protocol (iWARP) encapsulates RDMA semantics over standard TCP/IP. It uses:

- **RDMAP** (Remote Direct Memory Access Protocol) – the RDMA operation layer
- **DDP** (Direct Data Placement) – for zero-copy data delivery
- **MPA** (Marker PDU Aligned Framing) – for TCP stream alignment

iWARP is fully TCP-compatible and works on standard Ethernet infrastructure without requiring lossless fabrics. However, TCP’s complexity imposes higher NIC silicon cost, and iWARP has seen limited deployment compared to RoCE.

3.3 RoCE

RDMA over Converged Ethernet (RoCE) is the industry-dominant choice for deploying RDMA on Ethernet infrastructure. It is covered in detail in Section 4.

4 RoCE: RDMA over Converged Ethernet

4.1 Overview

RoCE was standardized by the InfiniBand Trade Association (IBTA) to bring RDMA capabilities to standard Ethernet networks. It preserves the InfiniBand Transport layer while replacing the IB network layer with Ethernet encapsulation. This allows organizations to leverage existing Ethernet investments while gaining RDMA performance.

4.2 RoCE v1

RoCE v1 (released 2010) encapsulates InfiniBand transport packets directly in Ethernet frames using EtherType 0x8915.

- Operates at **Layer 2** only – packets cannot be routed beyond a single broadcast domain (L2 subnet).
- Requires **Priority Flow Control (PFC)** on all switches in the path to achieve lossless delivery.
- Suitable for isolated, flat data center topologies.

4.3 RoCE v2

RoCE v2 (released 2014, also called “Routable RoCE” or “RRoCE”) adds UDP/IP encapsulation on top of the Ethernet frame:

$$\text{Ethernet} \supset \text{IP} \supset \text{UDP (port 4791)} \supset \text{IB BTH} \supset \text{Payload}$$

Key improvements over v1:

- **Layer 3 routable** – packets can traverse IP routers, enabling multi-subnet deployments and spine-leaf architectures.
- **ECMP support** – UDP source port hashing enables Equal-Cost Multi-Path load balancing across multiple links.
- **IPv4 and IPv6 support.**
- Retains the same requirement for lossless fabric (PFC + DCQCN).

RoCE v2 is the current industry standard and is implemented by all major RNIC vendors including NVIDIA (Mellanox), Broadcom, Marvell, and Intel.

4.4 Packet Format

Eth Header	IP Header	UDP Header	BTH	Payload	ICRC
14B	20B	8B	12B	Variable	4B

Table 2: RoCE v2 packet format. BTH = Base Transport Header; ICRC = Invariant CRC

The **Base Transport Header (BTH)** carries the Queue Pair Number (QPN), operation code (OPCODE), Packet Sequence Number (PSN), and other transport-layer fields inherited from InfiniBand.

4.5 Congestion Control: DCQCN

Because RDMA transports assume a lossless network (packet loss forces expensive re-transmissions that can stall the QP), congestion control is critical. The dominant algorithm for RoCE deployments is **DCQCN** (Data Center Quantized Congestion Notification), which combines:

- **ECN (Explicit Congestion Notification):** Switches mark packets when buffer occupancy exceeds a threshold using RED-like marking.
- **PFC (Priority Flow Control):** IEEE 802.1Qbb pause frames provide per-priority flow control as a last resort, halting transmission on a specific traffic class.
- **Rate-based control at the NIC:** Senders react to CNP (Congestion Notification Packets) generated by the receiver upon seeing ECN-marked packets, reducing their injection rate.

PFC alone is insufficient – it can cause head-of-line blocking and congestion spreading (“PFC storms”) if not paired with ECN-based rate control. Proper deployment requires careful QoS configuration across the entire fabric.

4.6 Lossless Fabric Requirements

Deploying RoCE at scale requires:

- PFC enabled on RDMA traffic class on all network ports
- ECN marking configured on all switch ports (typically using WRED)
- DSCP marking on RoCE traffic to enforce QoS policies
- Sufficient buffer space on switches (deep-buffer switches preferred)
- Careful network design to minimize PFC pause propagation

5 Performance Characteristics

5.1 Latency

RDMA’s kernel-bypass architecture eliminates the software overhead that dominates traditional TCP/IP latency. Typical latency figures:

Technology	One-way Latency	CPU Overhead
TCP/IP (standard)	50–200 μs	High (kernel involvement)
TCP/IP (kernel bypass, DPDK)	5–20 μs	Medium
RoCE v2 SEND/RECV	1–3 μs	Low (RNIC-driven)
RoCE v2 WRITE/READ	< 1 μs	Minimal
InfiniBand HDR	$\approx 0.6 \mu s$	Minimal

Table 3: Approximate one-way latency comparison for small messages

5.2 Throughput

Modern RoCE NICs (e.g., NVIDIA ConnectX-7) operate at 200 Gb/s or 400 Gb/s per port and can sustain near line-rate throughput for large transfers, with CPU utilization well below 10% — compared to near-100% CPU saturation for equivalent TCP bandwidth.

5.3 CPU Offload

The CPU benefits of RDMA are substantial:

- Zero data copies between kernel and user space
- No system calls during the data path (memory-mapped doorbells)
- Interrupt coalescing and polling modes
- The NIC handles segmentation, reassembly, and reliability

6 Software Ecosystem

6.1 Linux RDMA Stack

The Linux RDMA stack (`rdma-core`) provides:

- `libibverbs` – user-space Verbs API library
- `librdmacm` – RDMA Connection Manager (CM) for connection setup
- `ibv_*` utilities – diagnostic and benchmarking tools
- Kernel modules: `ib_core`, `rdma_ucm`, `mlx5_ib` (Mellanox), etc.

6.2 Key Benchmarking Tools

Listing 2: RDMA benchmarking with `perftest`

```

1  # Server side: RDMA Write bandwidth test
2  ib_write_bw -d mlx5_0
3
4  # Client side: connect and run test
5  ib_write_bw -d mlx5_0 <server_ip>
6
7  # Latency test
8  ib_send_lat -d mlx5_0 <server_ip>
```

6.3 Higher-Level Frameworks

UCX (Unified Communication X)

A portable communication framework supporting RDMA, TCP, and shared memory transports. Used by OpenMPI, NCCL, and others.

OpenMPI / MVAPICH2

MPI implementations with native RDMA support for HPC workloads.

NCCL (NVIDIA Collective Communications Library)

Uses RoCE/InfiniBand for AllReduce and other collective operations in distributed deep learning.

NVMe-oF Block storage protocol over RDMA (or TCP), enabling remote NVMe drives with near-local latency.

SPDK (Storage Performance Development Kit)

User-space storage stack using RDMA for ultra-low latency NVMe-oF targets.

7 Deployment Considerations

7.1 Network Topology

For RoCE deployments, **leaf-spine** (Clos) topologies are standard:

- Each server connects to one or more leaf switches
- Leaf switches interconnect via spine switches
- ECMP and RoCE v2 UDP hashing distribute flows across spine links
- Over-subscription ratios should be minimized for RDMA-heavy traffic (1:1 or 2:1 preferred)

7.2 Operational Challenges

PFC Storms Misconfigured PFC can cause pause frames to cascade through the fabric, halting large portions of the network. Watchdog timers and careful per-hop PFC domain design mitigate this risk.

Mixed Traffic Sharing lossless RDMA traffic with regular TCP traffic requires strict QoS separation (separate traffic classes/queues).

MTU Tuning RDMA performs best with 4096B or 9000B (jumbo frames) MTU; consistent MTU across all path devices is mandatory.

Memory Registration

Registering large memory regions can be slow. Hugepages and memory registration caches (ODP – On-Demand Paging) address this.

7.3 Security

RDMA's zero-copy nature introduces security considerations:

- Memory regions must be explicitly registered and protected via Protection Domains and R_Keys (remote keys)
- Network-level encryption (IPsec, MACSec) can be applied but may reduce performance
- Physical network isolation (VLANs, dedicated fabrics) is common in sensitive deployments

Feature	InfiniBand	RoCE v2	iWARP
Network Layer	IB	Ethernet/IP/UDP	Ethernet/IP/TCP
Routable	Limited	Yes (L3)	Yes (L3)
Lossless Fabric Required	Yes (native)	Yes (PFC+ECN)	No
Latency	$\sim 0.6 \mu\text{s}$	$\sim 1\text{--}2 \mu\text{s}$	$\sim 2\text{--}5 \mu\text{s}$
Infrastructure Cost	High (dedicated)	Medium (Ethernet)	Low (standard)
Industry Adoption	HPC, AI	Cloud, AI, HPC	Storage, NAS
Max Link Speed	400 Gb/s+	400 Gb/s+	100 Gb/s

Table 4: Comparison of RDMA transport technologies

8 Industry Adoption and Trends

8.1 Hyperscale Cloud

Major cloud providers have deployed RoCE at massive scale:

- **Microsoft Azure** uses RoCE (with DCQCN) as the foundation of its SmartNIC-based storage backend (Azure Storage) and RDMA for VM-to-VM communication.
- **Meta (Facebook)** deployed RoCE-based RDMA for its AI training clusters (using custom ASICs and the Wedge/Minipack switch platforms).
- **Google** uses a proprietary RDMA-like transport (Swift congestion control) across its Jupiter network fabric.
- **AWS** leverages EFA (Elastic Fabric Adapter) – a custom RDMA-capable network interface – for HPC and ML workloads on EC2.

8.2 AI Training

The explosion of large language model (LLM) training has driven unprecedented demand for RDMA infrastructure. Training a model like GPT-4 or Llama 3 requires tight synchronization across thousands of GPUs, where AllReduce communication latency directly impacts GPU utilization. RoCE (particularly over 400 Gb/s HDR/NDR InfiniBand or 400GbE) has become a standard requirement for AI training clusters.

8.3 NVMe over Fabrics

NVMe-oF with RoCE transport is increasingly deployed in all-flash storage arrays and disaggregated storage architectures, delivering storage latencies of 100–200 μs end-to-end, comparable to locally attached NVMe.

9 Conclusion

RDMA and RoCE represent a fundamental shift in how high-performance networked systems are designed. By offloading data transfer to the NIC and bypassing the OS

kernel, RDMA enables latencies and throughputs that approach hardware limits, while freeing CPU resources for application logic.

RoCE v2 has emerged as the practical standard for deploying RDMA over modern Ethernet infrastructure, offering L3 routability and compatibility with standard switch silicon. While it requires careful fabric engineering — particularly around PFC and congestion control — its performance benefits justify the operational complexity in latency-sensitive, bandwidth-intensive workloads.

As AI training clusters, disaggregated storage, and real-time analytics continue to scale, RDMA and RoCE will remain central to data center networking. Emerging developments such as **400G/800G Ethernet**, **SmartNICs/DPUs** (Data Processing Units), and **Ultra Ethernet Consortium (UEC)** standards will further evolve the RDMA ecosystem, addressing current limitations in lossless fabric management and bringing RDMA semantics to a broader range of deployments.

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